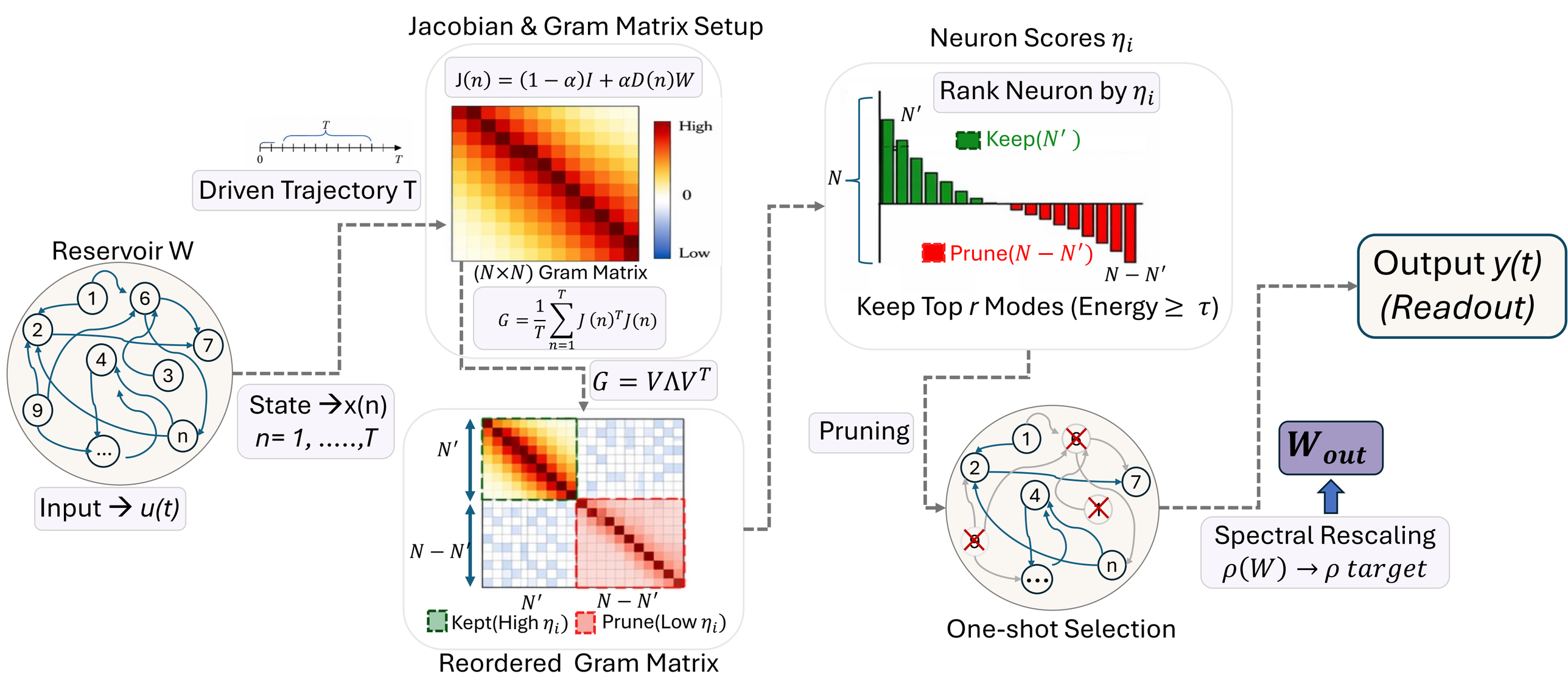

Jacobian & Gram Matrix Setup
$J(n) = (1-\alpha)I + \alpha D(n)W$
Driven Trajectory T
T
0
T
High
0
Low
$(N \times N)$ Gram Matrix
$G = \frac{1}{T}\sum_{n=1}^{T} J(n)^T J(n)$
Reservoir W
1
2
3
4
6
7
9
n
...
State →x(n)
n= 1, ...,T
Input → u(t)
$G = V\Lambda V^T$
N'
$N - N'$
N'
$N - N'$
Kept(High η_i)
Prune(Low η_i)
Reordered Gram Matrix
Neuron Scores η_i
Rank Neuron by η_i
N
N'
Keep(N')
Prune($N - N'$)
$N - N'$
Keep Top r Modes (Energy ≥ τ)
Pruning
One-shot Selection
Output y(t)
(Readout)
W_{out}
Spectral Rescaling
$\rho(W) \rightarrow \rho\ target$